AP CALCULUS AB

Unit 1 • Lesson 1.16 notes

Lesson 1.16 • Working with the Intermediate Value Theorem

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

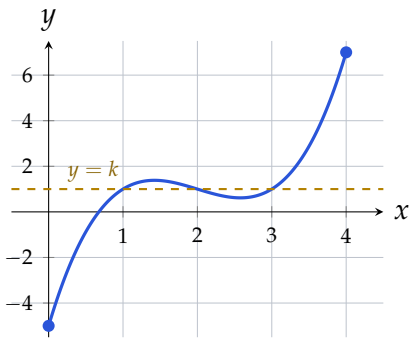
Learning goals

- State the Intermediate Value Theorem and its hypothesis.
- Use the IVT to justify the existence of solutions, with a complete AP-style justification.

Vocabulary. Intermediate Value Theorem (IVT), hypothesis, conclusion, existence.

Intermediate Value Theorem

If f is continuous on $[a, b]$ and k is a number between $f(a)$ and $f(b)$, then $f(c) = k$ for at least one c in $[a, b]$.



The IVT guarantees at least one c; here the line $y = 1$ meets the graph three times.

Example 1 • Existence of a root

Show that $x^3 + 2x = 5$ has a solution in $[1, 2]$.

Solution. Let $f(x) = x^3 + 2x$, a polynomial, continuous on $[1, 2]$. $f(1) = 3$ and $f(2) = 12$. Since $3 < 5 < 12$, the IVT guarantees a c in $[1, 2]$ with $f(c) = 5$.

Example 2 • Data in a table

A continuous function W has $W(0) = 8$, $W(2) = 3$, $W(5) = 6$, $W(9) = 1$. How many solutions of $W(t) = 4$ must there be on $[0, 9]$?

Solution. 4 is between 8 and 3 (on $[0, 2]$), between 3 and 6 (on $[2, 5]$), and between 6 and 1 (on $[5, 9]$). At least three.

Example 3 • When the IVT does not apply

$g(x) = \frac{1}{x}$ on $[-1, 1]$ has $g(-1) = -1$ and $g(1) = 1$, but never equals 0. No contradiction: g is not continuous on $[-1, 1]$.

Common mistakes

- Omitting the continuity statement in a justification.
- Concluding the IVT gives *exactly* one solution, or tells you where it is.

AP exam tip

Full-credit template: "f is continuous on $[a,b]$ because …. $f(a) = \ldots$ and $f(b) = \ldots$. Since k is between $f(a)$ and $f(b)$, by the Intermediate Value Theorem there is a c in $[a,b]$ with $f(c) = k$."

Quick check

1. Does $\cos x = x$ have a solution in $[0,1]$? Justify.
2. f is continuous, $f(1) = 4$, $f(3) = 4$. Does the IVT guarantee $f(c) = 0$ for some c in $[1,3]$?

Answers. 1. Yes: $h(x) = \cos x - x$ is continuous, $h(0) = 1 > 0$, $h(1) \approx -0.46 < 0$. 2. No: 0 is not between 4 and 4.